Synopsis

IoT Air and Sound Pollution Monitoring System

Submitted in partial fulfilment of the requirements of the Btech Second year students of

Third Semester

in

Computer Science and Engineering (Data Science)

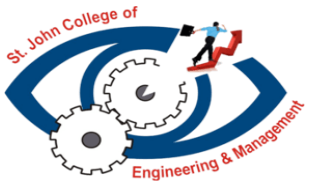
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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE) ST. JOHN COLLEGE OF ENGINEERING & MANAGEMENT

UNIVERSITY OF MUMBAI 2026-2027

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Acknowledgement

We express our sincere gratitude to all those who have directly and indirectly contributed to the successful preparation of this project synopsis.

We are deeply grateful to Dr. Kamal Shah, Principal, St. John College of Engineering and Management, for providing an excellent academic environment and the necessary facilities to undertake this project.

We extend our heartfelt thanks to Mrs. Joslyn Gracias e Vaz, Head of the Department of Computer Science and Engineering (Data Science), for her constant encouragement, valuable suggestions, and continuous support throughout the preparation of this synopsis.

We would like to express our sincere appreciation to our project guide, Prof. Sachin Sase, Assistant Professor, for his invaluable guidance, constructive suggestions, technical expertise, and continuous motivation. His encouragement and insightful feedback have been instrumental in shaping this project synopsis.

We also acknowledge all the faculty members and laboratory staff of the Department of Computer Science and Engineering (Data Science) for their timely assistance, cooperation, and technical support.

Finally, we express our heartfelt gratitude to our parents, family members, friends, and classmates for their constant encouragement, moral support, and cooperation throughout this work.

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ABSTRACT

Rapid urbanization, growing vehicular density, and unregulated industrial and construction activity have caused a steady rise in both air and noise pollution in Indian cities, posing serious risks to public health and the environment. Conventional pollution monitoring stations are few in number, expensive to set up, and provide only sparse, delayed readings that fail to capture localized pollution hotspots. This project proposes the design and development of a low-cost IoT Air and Sound Pollution Monitoring System that continuously senses key air quality parameters — including CO, CO2, smoke, and particulate matter (PM2.5/PM10) — along with ambient sound levels, temperature, and humidity, using an ESP32 microcontroller interfaced with the MQ-series gas sensors, a dust sensor, and a sound sensor module. The acquired data is processed on-board and displayed in real time on a local OLED display, with local buzzer/LED alerts triggered whenever pollutant or noise levels exceed permissible limits defined by the Central Pollution Control Board (CPCB). The entire unit is powered from a portable USB power bank, which keeps the design simple, compact, and easy to deploy at any location. The proposed system is compact, scalable, and significantly more economical than commercial-grade monitoring stations, making it suitable for deployment as a dense sensor network across residential colonies, traffic junctions, industrial zones, and educational campuses. The system is expected to deliver a working hardware prototype capable of real-time sensing, on-site data display, and threshold-based alerting, thereby supporting data-driven decisions for pollution mitigation and contributing towards the broader goals of smart and sustainable city infrastructure.

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Abbreviations and Symbols

- IoT: Internet of Things
- AI: Artificial Intelligence
- ML: Machine Learning
- CPCB: Central Pollution Control Board
- PM: Particulate Matter
- CO: Carbon Monoxide
- CO2: Carbon Dioxide
- dB: Decibel
- MCU: Microcontroller Unit
- UI: User Interface
- WBS: Work Breakdown Structure
- SDG: Sustainable Development Goal
- BOM: Bill of Materials

CHAPTER 1: INTRODUCTION

1.1 Background

Air and noise pollution have emerged as two of the most pressing environmental challenges faced by rapidly urbanizing Indian cities. According to reports of the Central Pollution Control Board (CPCB) and the World Health Organization (WHO), most metropolitan regions in India regularly record particulate matter (PM2.5 and PM10) and ambient noise levels well above prescribed safe limits, driven primarily by vehicular emissions, construction dust, industrial exhaust, and traffic congestion. Prolonged exposure to such pollutants is strongly associated with respiratory illnesses, cardiovascular disease, hearing impairment, and reduced quality of life. The Internet of Things (IoT) offers a practical, low-cost, and scalable approach to continuously sense, process, and display environmental data, enabling authorities, institutions, and citizens to make timely, data-driven decisions rather than relying on infrequent manual sampling.

1.2 Problem Statement

Existing pollution monitoring infrastructure in most Indian cities, including Mumbai, is sparse and centralized:

- Existing Problem: Government-run air quality monitoring stations are few in number and located mostly in select localities, leaving large residential, institutional, and industrial areas without any real-time pollution data.
- Industrial Issue: Commercial-grade monitoring equipment is expensive (often exceeding several lakhs of rupees per station), which discourages dense deployment across smaller campuses, housing societies, and traffic junctions.
- Research Gap: Most existing IoT-based systems studied in literature monitor either air quality or sound levels in isolation, and very few combine both parameters with real-time local alerting on a single, low-cost, self-contained hardware platform.
- Need for Improvement: There is a clear need for an affordable, compact, and easily replicable IoT device that can be deployed at multiple points to build a distributed, hyper-local pollution sensing network.

1.3 Need of the Project

A low-cost IoT air and sound pollution monitoring system directly addresses the gap between the high cost of government-grade monitoring stations and the growing need for hyper-local, real-time environmental data. Such a system empowers educational institutions, resident welfare associations, and local municipal bodies to independently track pollution trends in their immediate vicinity, issue timely alerts,

and generate historical data that can support both individual health precautions and long-term urban planning and policy decisions.

1.4 Objectives

General Objective: To design, develop, and test a low-cost, IoT-enabled hardware prototype capable of simultaneously monitoring air quality and sound pollution levels in real time and alerting users when thresholds are exceeded.

Specific Objectives:

- Design a compact sensor module integrating gas sensors (MQ-135, MQ-7), a dust/particulate sensor, a sound level sensor, and a temperature-humidity sensor around an ESP32 microcontroller, powered from a portable USB power bank.
- Develop firmware to acquire, filter, and process sensor data at defined sampling intervals.
- Develop an on-board display and alert interface (OLED display, buzzer, and LED) for real-time visualization of sensor readings at the point of measurement.
- Implement threshold-based local (buzzer/LED) alerting referenced against CPCB safe limits.
- Fabricate and test a working hardware prototype under real ambient conditions to validate sensor accuracy and system reliability.
- Evaluate the system's cost-effectiveness in comparison with existing commercial pollution monitoring solutions.

1.5 Scope

What will be covered:

- Real-time sensing of CO, CO2/smoke, particulate matter (PM2.5/PM10), ambient sound level (dB), temperature, and humidity.
- On-board data processing and real-time display of readings on a local OLED display.
- Threshold-based local alert generation (buzzer/LED).

Limitations:

- The system provides indicative, low-cost sensing rather than laboratory-grade, reference-instrument accuracy.
- Coverage is limited to the deployment points of the prototype units within the scope of this synopsis; a city-wide sensor mesh is beyond the current scope.

Future expansion:

- Integration of machine learning models for pollution forecasting and anomaly detection.
- GPS-tagged mobile monitoring units and a city-wide sensor mesh network.

1.6 Expected Outcomes

- A working IoT hardware prototype for combined air and sound pollution monitoring.
- Companion firmware and an on-board OLED display with buzzer/LED alerts for real-time visualization of sensor readings.
- A validated, low-cost bill of materials demonstrating feasibility for large-scale, distributed deployment.
- A technical report/publication documenting design methodology, testing, and results.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This chapter reviews recent research on IoT-based air and sound pollution monitoring systems to identify prevailing design approaches, sensing technologies, and open research gaps that this project seeks to address.

2.2 Literature Review Table

Author(s)	Year	Method	Findings	Research Gap
Alam et al.	2019	ESP8266 + MQ sensors, dust sensor, sound sensor	Low-cost combined gas, sound, dust and temperature monitoring with web connectivity	Limited calibration accuracy across varying humidity conditions
Deshmukh, Surendran & Sardey	2017	Distributed sensor nodes with central server architecture	Cost-efficient conceptual IoT architecture for combined air-sound monitoring	No real-time mobile alerting mechanism implemented
Gupta et al.	2019	IoT sensor nodes for smart city air quality	Effective localized air pollution monitoring for smart cities	Sound pollution not addressed
Aserkar et al.	2024	IoT-enabled optical sensors	High-precision realtime air quality monitoring in smart cities	Higher hardware cost due to optical sensing modules
Ali, A.	2022	IoT + smart sensor framework	Scalable smart-city air pollution monitoring framework	Framework-level study; limited hardware validation
Herath et al.	2022	IoT-based realtime air purification system	Combines monitoring with active outdoor air purification	Increased system complexity and power requirements

Ara, F.	2023	Microcontroller + multi-sensor array + cloud	Real-time combined air and sound monitoring with cloud analytics	Limited longterm field deployment testing reported

Author(s)	Year	Method	Findings	Research Gap
Chen, Liu & Xu	2015	WSN-based forecasting model	Early IoT-based air pollution monitoring and forecasting system	Forecasting accuracy limited by sparse historical data
Liu et al.	2012	Wireless sensor network nodes	Urban air quality monitoring using WSN technology	High deployment and maintenance cost for large WSN
Subahi & Bouazza	2020	IoT-based greenhouse monitoring system	Intelligent monitoring and control applicable to environmental sensing	Domain-specific to greenhouse, not urban pollution
Liu, Wang, Zhao & Yang	2021	Multi-point indoor IoT sensor network	Indoor air quality monitoring at multiple points using IoT	Focused on indoor environments only
Joshi, L.M.	2017	Arduino-based gas and sound sensors	Combined air and sound pollution monitoring on a single low-cost platform	Limited cloudbased data logging and analytics
Nishant et al.	2021	Multiparameter IoT sensor system	Industrial air, water, and noise pollution monitoring	System complexity increases with added waterquality module

Waghmare et al.	2019	IoT-based sensor and controller unit	Monitoring and controlling of air pollution using IoT	Control mechanism tested only in a limited lab setup
Okokpujie et al.	2018	Smart sensorbased monitoring node	Smart, low-cost air pollution monitoring system	Sound pollution parameter not included
Pawar, Shinde & Sharma	2023	Arduino + GPS + Wi-Fi sensor kit	Low-cost, GPStagged pollution monitoring kit	GPS module adds cost and power overhead

2.3 Research Gap

Prior work has largely treated air quality monitoring and noise/sound pollution monitoring as separate problems, with only a limited number of studies (Alam et al., 2019; Deshmukh et al., 2017; Ara, 2023; Joshi, 2017) attempting to combine both on a single platform. Among these combined systems, most either depend on network connectivity and external servers, or lack rigorous field validation against CPCB threshold limits. Additionally, several systems reviewed rely on costlier optical or WSN-based hardware that limits scalability for dense, low-cost deployment. This project addresses these gaps by combining multi-parameter air and sound sensing, low-cost ESP32-based processing, on-board OLED display of readings, and threshold-based local (buzzer/LED) alerting into a single, replicable, and economically deployable unit powered by a portable USB power bank.

2.4 Novelty / Innovation

- Unified sensing of gaseous pollutants, particulate matter, and ambient sound on a single compact, low-cost hardware unit.
- Local alerting: immediate on-site buzzer/LED indication together with a real-time OLED display of readings, without dependence on any network.
- Design optimized for low bill-of-materials cost, enabling dense multi-point deployment rather than a handful of expensive stations.
- Standalone, portable operation from a USB power bank, enabling deployment at locations without Wi-Fi coverage or mains power.

CHAPTER 3: PROPOSED METHODOLOGY

3.1 Proposed System

The proposed system is a compact, power-bank-powered IoT unit built around an ESP32 microcontroller that interfaces with gas, dust, sound, temperature, and humidity sensors. Sensor readings are sampled at regular intervals, and processed on-board. The unit is powered from a portable USB power bank, while the local OLED display, buzzer, and LED indicators show the live readings and give immediate on-site feedback when readings cross CPCB-defined safe thresholds.

3.2 Working Principle

1. The MQ-135, MQ-7, dust, and sound sensors continuously sense their respective environmental parameters and output analog/digital voltage signals proportional to pollutant concentration or noise intensity.
2. The ESP32 microcontroller reads these signals through its ADC and digital I/O pins at a fixed sampling interval.
3. Raw readings are filtered (moving-average smoothing) and converted into calibrated units (ppm for gases, µg/m³ for particulates, dB for sound).
4. Processed values are compared against pre-defined CPCB safe-limit thresholds.
5. If a threshold is breached, the buzzer and LED are triggered locally to alert people nearby.
6. All readings, whether within or beyond safe limits, are shown on the local OLED display in real time.
7. The unit runs continuously from a 5V USB power bank, and the display is refreshed at every sampling interval so that users can track pollution levels on site.

3.3 System Architecture / Block Diagram

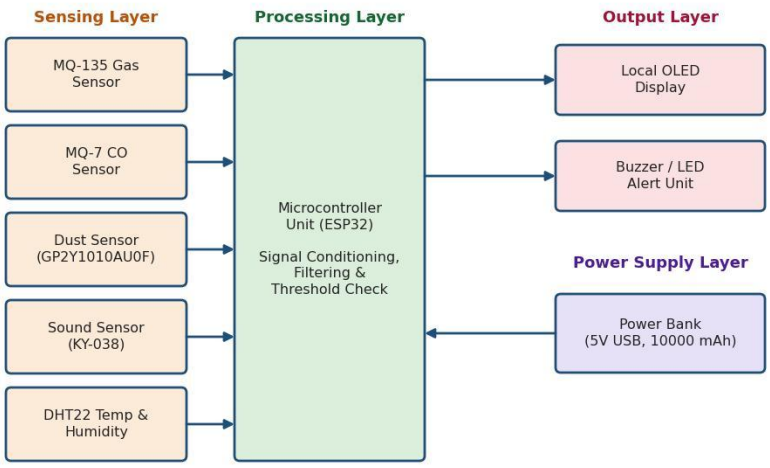


Figure 1: System Architecture / Block Diagram 3.4

Flow Chart

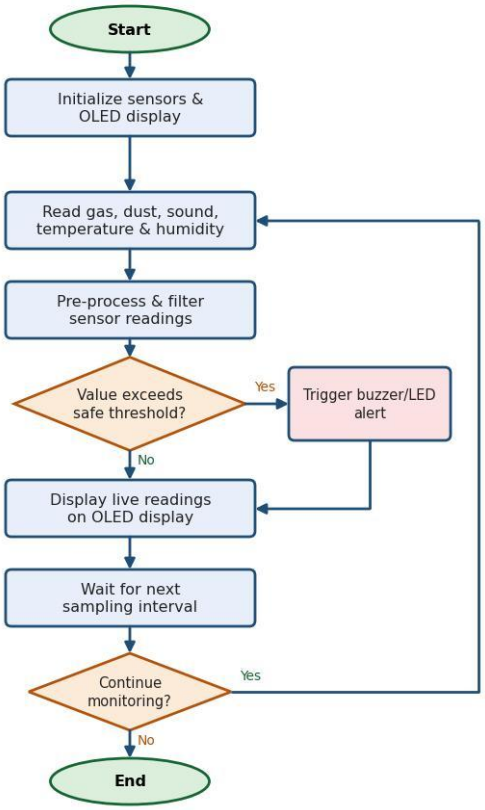


Figure 2: System Flowchart

3.5 Methodology

Design procedure: The hardware design begins with sensor selection based on target pollutants and required accuracy, followed by circuit design and PCB layout for compact integration. Experimental procedure: Each sensor is individually calibrated against known reference values before integration into the full system. Algorithms: A moving-average filter is applied to smooth noisy analog readings, and a rule-based threshold comparison algorithm triggers alerts; a future extension may incorporate a lightweight regression model for short-term pollution trend prediction. Manufacturing process: The prototype is assembled on a perfboard/PCB and housed in a compact, ventilated enclosure suitable for outdoor or semi-outdoor deployment. Programming steps: Firmware is developed in the Arduino IDE/ESP-IDF environment in C/C++, covering sensor driver initialization, threshold-based alert logic, and OLED display rendering. **3.6 Components / Materials**

Component	Specification	Quantity

ESP32 Development Board	Dual-core microcontroller, 240 MHz	1
MQ-135 Gas Sensor	Detects NH3, NOx, CO2, benzene, smoke	1
MQ-7 CO Sensor	Carbon monoxide detection, 20–2000 ppm	1
GP2Y1010AU0F Dust Sensor	Optical PM2.5/PM10 particulate sensor	1
KY-038 Sound Sensor Module	Microphone-based ambient sound level sensor	1
DHT22 Sensor	Temperature & humidity sensor, ±0.5°C accuracy	1
0.96" OLED Display	I2C local readout display	1
Active Buzzer + LED Indicators	Local threshold-breach alert	1 set
Power Bank (USB) 10000 mAh	Portable 5V USB power supply	1
PCB / Perfboard & Enclosure	Mounting and weatherproof housing	1
Connecting Wires, Resistors & Breadboard	Circuit assembly & prototyping	As required

CHAPTER 4: PROJECT PLANNING

4.1 Work Breakdown Structure

S. No.	Task
1	Requirement analysis & literature survey
2	Component selection & procurement
3	Circuit design & sensor interfacing
4	Firmware development
5	Power supply integration & display/alert interfacing
6	Hardware assembly & enclosure fabrication
7	Sensor calibration & unit testing
8	Integration testing & field trial
9	Documentation & report preparation

4.2 Gantt Chart (Semester Planning)

Task	Wk 1-2	Wk 3-4	Wk 5-6	Wk 7-8	Wk 9-10	Wk 11-12
Requirement Analysis & Literature Survey	✓					
Component Selection & Procurement	✓	✓				
Circuit Design & Sensor Interfacing		✓	✓			
Firmware Development			✓	✓		
Power Supply & Display Integration				✓	✓	

Hardware Assembly & Testing				✓	✓	
Integration & Field Trial					✓	✓
Documentation & Report						✓

Figure 3: Gantt Chart – Semester Planning

4.3 Roles and Responsibilities

Student	Responsibility
Aditya Mishra	Hardware design, sensor interfacing & circuit integration
Rohit Jadhvar	Firmware development & microcontroller programming
Aryan Chaubal	Power bank integration, OLED display & buzzer/LED alert interfacing
Pranab Ghorai	Testing, calibration, documentation & report compilation

4.4 Risk Assessment

Risk	Impact	Mitigation
Sensor inaccuracy due to lack of calibration	Medium	Calibrate each sensor against known reference sources before deployment
Power bank auto shut-off at low current draw	Medium	Use a power bank with an always-on/low-current mode, or add a small dummy load
Power supply interruption	Low	Use a fully charged power bank of adequate capacity and keep a spare unit ready
Component procurement delay	Low	Order components with buffer lead time and identify alternate vendors

Environmental damage to sensors (dust/moisture)	Medium	Use a ventilated, weatherproof enclosure for outdoor units

CHAPTER 5: RESOURCES & FEASIBILITY

5.1 Technical Feasibility

All hardware components (ESP32, MQ-series gas sensors, dust sensor, sound sensor, DHT22) are widely available, well documented, and compatible with open-source development tools such as the Arduino IDE. The processing requirements are well within the capability of the ESP32's dual-core processor, and the whole unit can be powered from a standard 5V USB power bank, making the system technically feasible to design, assemble, and test within the project timeline.

5.2 Economic Feasibility

The total estimated hardware cost of a single monitoring unit is under ₹3,000, substantially lower than commercial pollution-monitoring stations that can cost several lakhs of rupees. This low per-unit cost makes it economically feasible to deploy multiple units for a distributed sensor network within a modest budget.

5.3 Operational Feasibility

The system is designed for simple installation with minimal maintenance, requiring only periodic sensor calibration checks and power bank recharging. The OLED display and buzzer/LED indicators require no specialized training to interpret, making the system operationally feasible for deployment by institutional staff or municipal personnel without deep technical expertise.

5.4 Budget Estimation

Item	Quantity	Cost
ESP32 Development Board	1	₹450
MQ-135 Gas Sensor	1	₹150
MQ-7 CO Sensor	1	₹180
GP2Y1010AU0F Dust Sensor	1	₹450
KY-038 Sound Sensor Module	1	₹80
DHT22 Temp & Humidity Sensor	1	₹350
0.96" OLED Display	1	₹250

Buzzer & LEDs	1 set	₹50
Power Bank (5V, 10000 mAh)	1	₹400
Enclosure, PCB/Perfboard, Wires & Misc.	—	₹600

Estimated Total: ₹2,960 (per unit, approximate) 5.5

Bill of Materials (BOM)

Item	Quantity	Cost
ESP32 Development Board	1	₹450
MQ-135 Gas Sensor	1	₹150
MQ-7 CO Sensor	1	₹180
GP2Y1010AU0F Dust Sensor	1	₹450
KY-038 Sound Sensor Module	1	₹80
DHT22 Temp & Humidity Sensor	1	₹350
0.96" OLED Display	1	₹250
Buzzer & LEDs	1 set	₹50
Power Bank (5V, 10000 mAh)	1	₹400
Enclosure, PCB/Perfboard, Wires & Misc.	—	₹600

CHAPTER 6: EXPECTED RESULTS

The following outcomes are expected upon completion of the project (no actual experimental results are presented at the synopsis stage):

- A functional prototype capable of simultaneously sensing air pollutants (CO, CO2/smoke, PM2.5/PM10) and ambient sound levels.
- Expected sensing accuracy within ±10-15% of reference-grade instruments for gas and particulate readings, acceptable for indicative, low-cost monitoring.
- Expected system response and alert-trigger latency of under 2–3 seconds from threshold breach to local notification (buzzer/LED indication and display update).
- Reliable continuous operation and on-site monitoring over extended field trial durations.
- A functioning on-board OLED display capable of showing live readings of all monitored parameters.
- Stable operation of the complete unit when powered from a USB power bank, with continuous run time depending on the power bank capacity.

CHAPTER 7: SUSTAINABILITY & IMPACT

7.1 Industrial Relevance

The proposed system's low-cost, modular design makes it directly relevant to industries seeking affordable environmental compliance monitoring, as well as to smart-city infrastructure providers and facility-management companies that require distributed, real-time pollution sensing at scale.

7.2 Societal Impact

By making real-time pollution data accessible to citizens, educational institutions, and local governing bodies, the system supports informed personal health decisions (e.g., avoiding high-pollution areas or times) and encourages community-level awareness and advocacy for cleaner air and quieter surroundings.

7.3 Environmental Impact

The system itself is low-power and portable, running from a rechargeable USB power bank, and its core purpose — continuous, dense environmental monitoring — directly supports environmental protection by enabling earlier detection of pollution spikes, informing targeted corrective action, and reducing the cumulative health and ecological impact of unchecked air and noise pollution.

7.4 SDG Mapping

- SDG 3 – Good Health and Well-being
- SDG 9 – Industry, Innovation and Infrastructure
- SDG 11 – Sustainable Cities and Communities
- SDG 13 – Climate Action

CHAPTER 8: CONCLUSION

Rising air and sound pollution levels in urban areas pose serious risks to public health, and the sparse, costly nature of existing government monitoring infrastructure leaves a significant gap in localized, real-time pollution data. This project proposes a lowcost, ESP32-based IoT system that combines multi-parameter air quality sensing (CO, CO2/smoke, PM2.5/PM10) with ambient sound level monitoring, temperature, and humidity, coupled with on-board OLED display of readings and local buzzer/LED threshold alerting, all powered from a portable USB power bank. The proposed solution is expected to deliver a compact, affordable, and easily replicable hardware prototype suitable for dense, distributed deployment across residential, institutional, and industrial zones. By enabling continuous, accessible pollution monitoring at a fraction of the cost of commercial-grade stations, this project is expected to contribute meaningfully toward data-driven pollution mitigation, public health awareness, and the broader objectives of smart, sustainable urban infrastructure.

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APPENDIX

The following supporting material will be compiled during project execution and attached in the final report:

- Detailed circuit / schematic diagram of sensor-to-microcontroller connections
- Firmware source code listing (Arduino/ESP-IDF)
- CPCB reference threshold tables used for calibration
- Data sheets of key components (ESP32, MQ-135, MQ-7, GP2Y1010AU0F, DHT22)